

Machine Learning

Detecting Points of Interest (PoI) Using Clustering Algorithms



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Abstract

This report presents a study on the detection of points of interest (PoI) in Lyon using clustering algorithms and a Flickr posts dataset. The objective is to identify geographically significant locations using spatial data and to evaluate various clustering techniques.

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1 Introduction

The goal of this project is to identify significant points of interest in Lyon using geospatial data and clustering techniques. By analyzing data from Flickr, a popular photo-sharing platform, we aim to identify clusters of geographic points that may represent notable locations in the city. Users on Flickr often geotag their photos, providing valuable spatial data that can be used to identify areas of interest. We will explore the application of several clustering algorithms to determine which is most suitable for this task.

2 Dataset Description

The dataset used in this study contains geolocated Flickr posts. It includes 83,837 entries and 16 different features. The features in this dataset are as follows:

Feature Name	Description
id	A unique identifier for each post.
user	The user who posted the image
lat	The latitude of the post's geolocation.
long	The longitude of the post's geolocation.
tags	Tags associated with each image,
title	The title given by the user to the post.
date_taken_minute	The minute when the photo was taken.
date_taken_hour	The hour when the photo was taken.
date_taken_day	The day when the photo was taken.
date_taken_month	The month when the photo was taken.
date_taken_year	The year when the photo was taken.
date_upload_minute	The minute when the photo was uploaded.
date_upload_hour	The hour when the photo was uploaded.
date_upload_day	The day when the photo was uploaded.
date_upload_month	The month when the photo was uploaded.
date_upload_year	The year when the photo was uploaded.

Table 1: Description of the dataset features

3 Preprocessing

The preprocessing steps include several important stages to prepare the dataset for analysis by clustering algorithms:

- **Renaming columns:** The dataset may have columns with extra spaces or misnamed attributes. For example, the latitude column is initially labeled as `_lat`, which can create inconsistencies. We strip any leading or trailing spaces and ensure that all column names are standardized. For instance, `_lat` is renamed to `lat`.
- **Removing duplicates:** To ensure the dataset is not inflated with repetitive entries, we check for duplicate rows. We identify and drop duplicates to retain only unique entries.
- **Dropping irrelevant columns:** For the scope of this analysis, we focus only on location-related data. Other features such as tags or dates are irrelevant for this task. Thus, we drop columns that do not contribute to the analysis, leaving only the location data, i.e., the `lat` (latitude) and `long` (longitude) columns.
- **Checking for missing values:** To ensure data completeness, we verify that there are no missing or null values in the dataset.

After completing the preprocessing steps, we reduced the dataset from approximately 83,000 posts to around 15,000 entries. Despite the reduction in the dataset, we noticed that approximately 8,000 of the remaining entries are geographical duplicates. This means that these posts are located at the exact same geographic coordinates (`lat`, `long`).

In most analysis contexts, duplicates like these might be removed to avoid redundancy. However, for our particular case, retaining these entries is crucial. The presence of geographical duplicates carries valuable information that will be leveraged in our Point of Interest (PoI) detection process. Since the goal is to detect clusters of points in space, these duplicate entries are indicative of high-density locations where multiple posts are associated with the same geographic point.

4 Data Visualization

4.1 Initial Visualization

In order to better understand the dataset, it is crucial to place these entries in their geographical context. To achieve this, we use the `folium` library, which allows us to overlay the data points on a map of Lyon. This visualization helps us to observe the spatial distribution of the points, providing a clearer understanding of the dataset’s geographical characteristics.

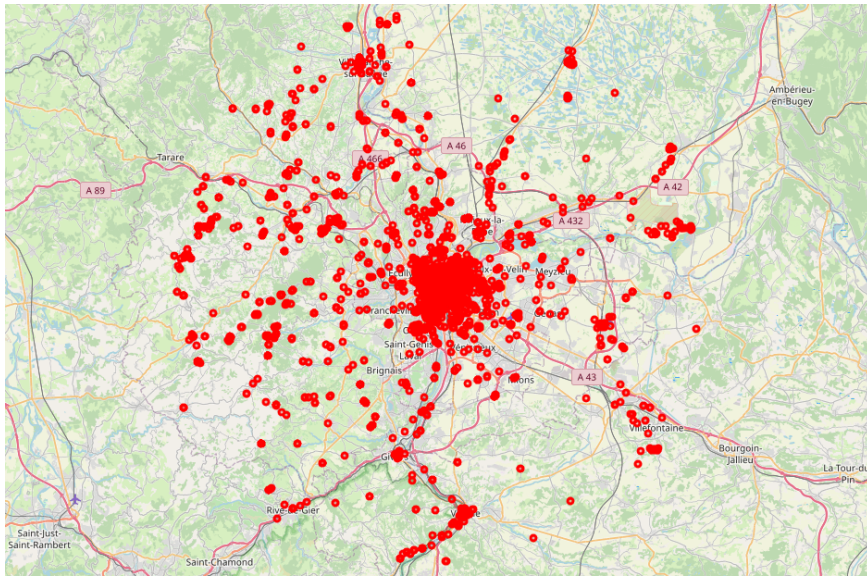


Figure 1: Scatter plot of geolocated posts in the dataset.

As observed from the initial visualization, the entire dataset spans a wide geographical area, making it challenging to identify meaningful clusters within the city of Lyon. The large geographical spread of data points creates significant spatial dispersion, which complicates the task of detecting localized points of interest (PoI). Therefore, to focus on areas with higher concentrations of data, we decided to create two subsets of the dataset.

The first subset includes only the posts located within the city limits of Lyon, which helps narrow down the focus to the central urban area. The second subset zooms in even further, isolating posts within the 1st, 2nd, 3rd, and 4th arrondissements, which encompass the Presqu'île and Vieux Lyon areas. These districts account for more than a third of the entire dataset, with approximately 6,000 entries, which should enable us to extract local points of interest (PoIs) with greater precision.

To illustrate these concentrations before applying clustering algorithms, we present two heatmaps below:

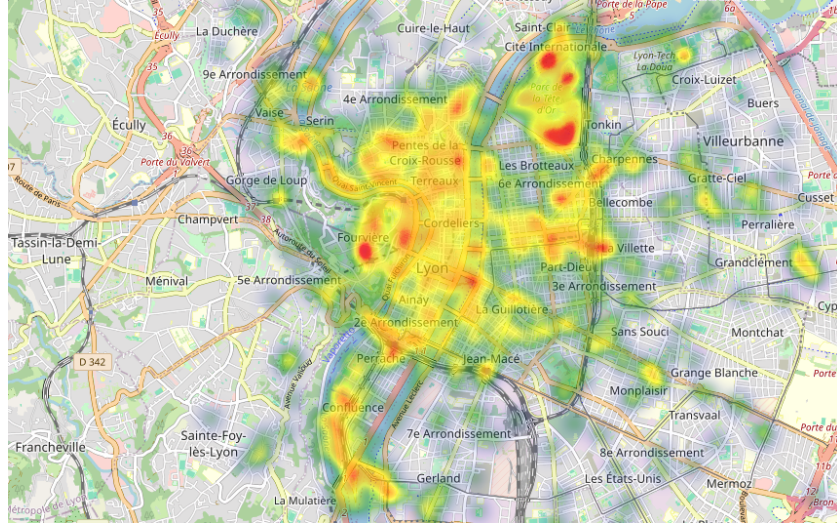


Figure 2: Heatmap of data density within the city of Lyon

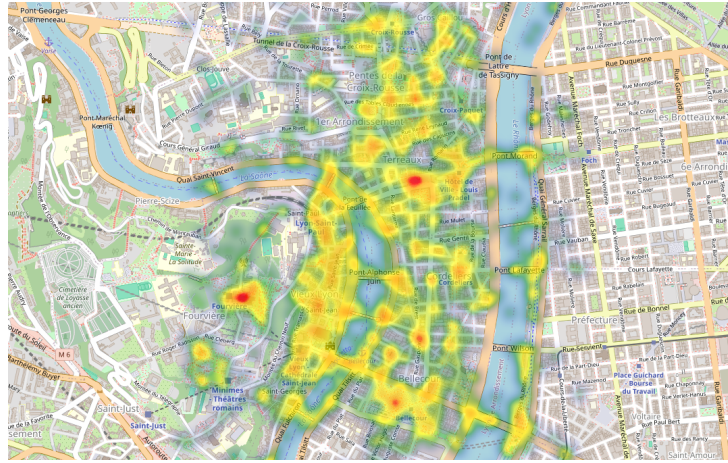


Figure 3: Heatmap of data density within central districts of Lyon

These heatmaps serve as a preliminary step in visualizing density variations before clustering methods are applied. By identifying high-density regions, we can better assess the effectiveness of different clustering algorithms in detecting PoIs.

5 Clustering Algorithms

The goal of clustering is to identify regions with high concentrations of Flickr posts, which can indicate the presence of significant PoIs. We apply and compare the following clustering algorithms:

- **K-Means:** A centroid-based algorithm that partitions data into a predefined number of clusters.
- **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise): A density-based algorithm that can detect clusters of arbitrary shapes without requiring the number of clusters to be specified.
- **Agglomerative Clustering:** A hierarchical clustering method that builds clusters by recursively merging or splitting them.
- **Mean-Shift Clustering:** A density-based algorithm that iteratively shifts a window towards regions of higher data density.

For each algorithm, the hyperparameters used will be provided to ensure reproducibility. These parameters have been fine-tuned manually to achieve the best clustering performance and the most meaningful associations.

5.1 K-Means Clustering

While this algorithm is efficient and easy to implement, it has notable limitations for spatial data. Specifically, K-Means assumes clusters are spherical and of similar density, which may not align with the natural distribution of geographic points of interest. Additionally, it requires specifying the number of clusters in advance, which can be challenging without prior knowledge of the data structure.

Given these constraints, we expect K-Means to provide only a rough approximation of the underlying clusters, likely producing suboptimal results. However, it will serve as an initial benchmark before moving on to more suitable algorithms.

5.1.1 Hyperparameters

For K-Means, we applied the following hyperparameters:

- `n_clusters = 10`
- `random_state = 42`

5.1.2 Results

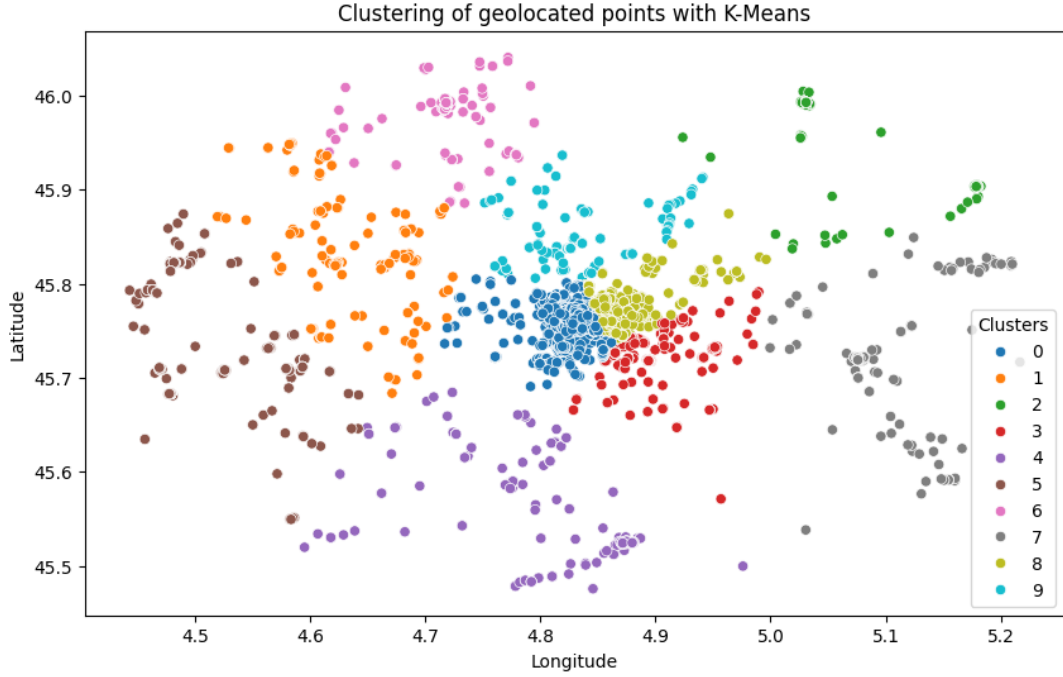


Figure 4: Clustering of geolocated points with K-Means

5.1.3 Conclusion

K-Means struggles to detect meaningful clusters in geospatial data due to its assumption of spherical clusters. The predefined number of clusters also limits the algorithm's flexibility in identifying natural groupings of points. As a result, K-Means provides a rough approximation rather than a reliable model for detecting PoIs in Lyon. Thus, we move forward with other algorithms that better handle geographic data.

5.2 DBSCAN

This algorithm appears to be the most suitable for our problem, as it can effectively detect clusters of geographic points of interest while filtering out noise. Its density-based approach allows for the identification of meaningful structures without requiring a predefined number of clusters, making it well-suited for spatial data analysis.

Since this algorithm proved to be highly effective, we applied it to our three datasets (twice for the most reduced one).

5.2.1 Hyperparameters

For DBSCAN, the following hyperparameters were used (resp. for each dataset):

- `eps` = 0.04, 0.005, 0.0009, 0.0005
- `min_samples` = 150, 200, 100, 40

5.2.2 Results

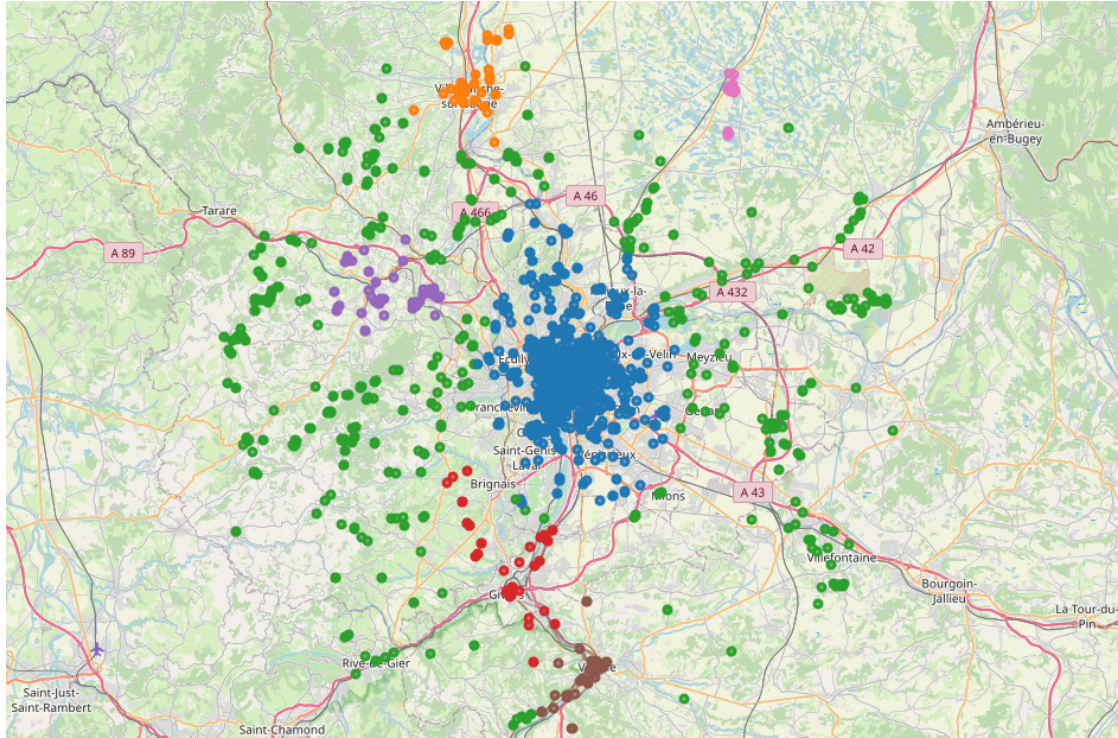


Figure 5: Clustering of geolocated points with DBSCAN

We can already identify five important clusters:

- **Lyon** (which we will detail later) - blue
- **Villefranche** (north of Lyon) - orange
- **Givors** (south of Lyon) - red
- **Vienne** (south of Givors) - brown
- **Parc des Oiseaux** near Villars-les-Dombes (northeast of Lyon) - pink

There remains one last cluster – purple – which is not very useful as it does not correspond to a single point of interest but rather to multiple locations such as Lentilly or the Couvent Sainte-Marie de la Tourette.

We next use the Lyon dataset :

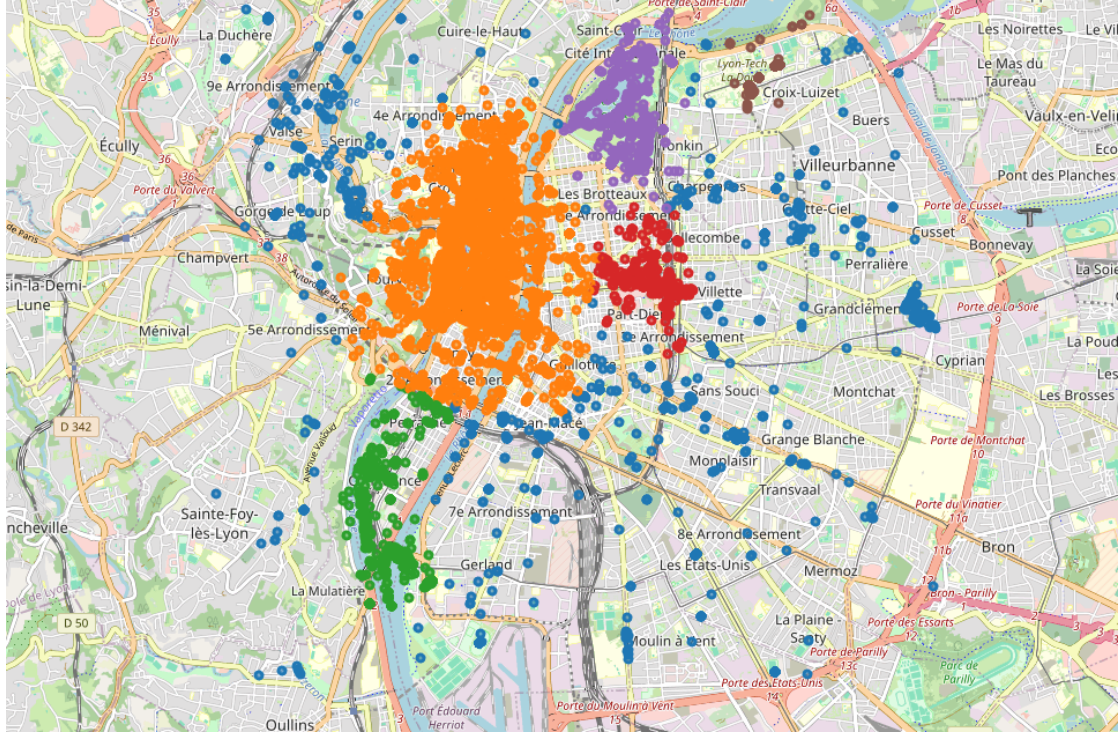


Figure 6: Clustering of geolocated points in Lyon with DBSCAN

For Lyon as a whole, we can identify five additional clusters:

- **Lyon city center** (including the 1st, 2nd, 3rd, and 4th districts) - orange
- **Confluence** (in the south) - green: We could focus on this area and divide it into two sub-clusters (the Saône riverbanks and the Musée des Confluences).
- **Parc de la Tête d'Or** - purple
- **La Part-Dieu** (shopping mall & railway station) - red
- **Lyon Tech La Doua** (especially INSA) - brown

We precise our PoI detection with the center of Lyon :

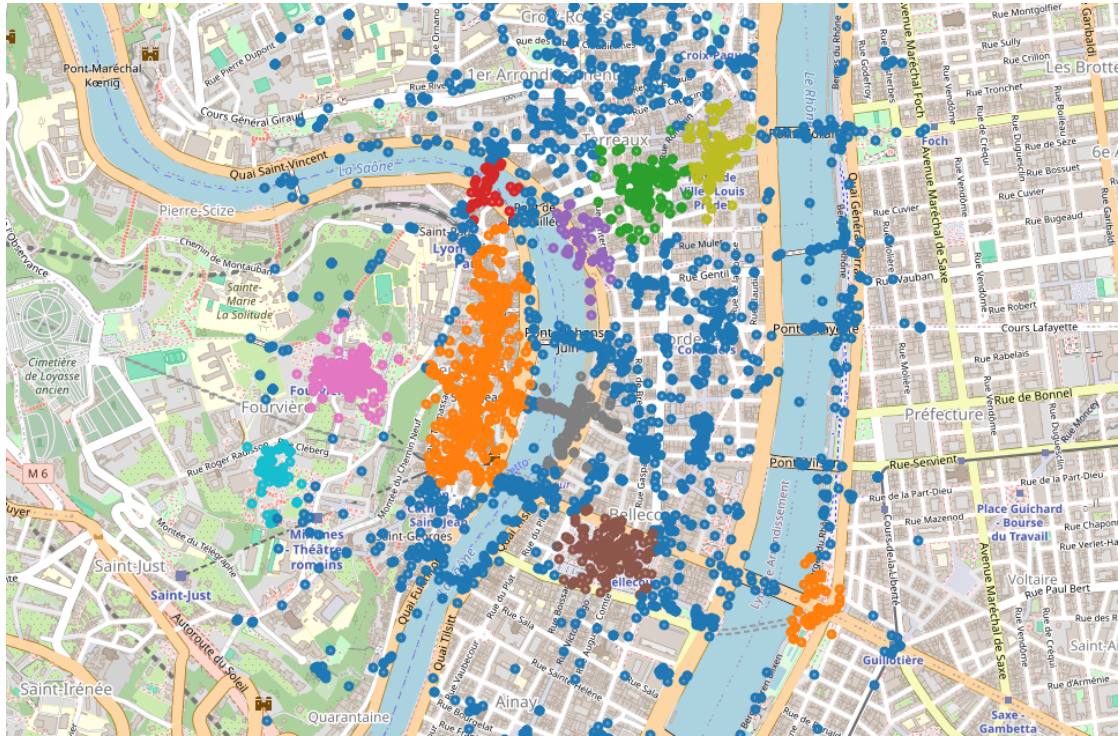


Figure 7: Clustering in the center of Lyon with DBSCAN (Main Clusters)

In this first clustering of the center of Lyon, we identify the main points of interest:

- **Basilica of Fourvière** - rose
- **Roman Theater** - cyan
- **Old Lyon** - orange
- **Saône riverbank opposite Old Lyon** - purple and grey
- **Place des Terreaux** - green
- **City Hall - Louis Pradel** - lime
- **Bellecour** - brown
- **Rhône riverbank (near Bellecour)** - orange

As an other exemple to get more precise PoI, we ran this algorithm with other hyperparameters:

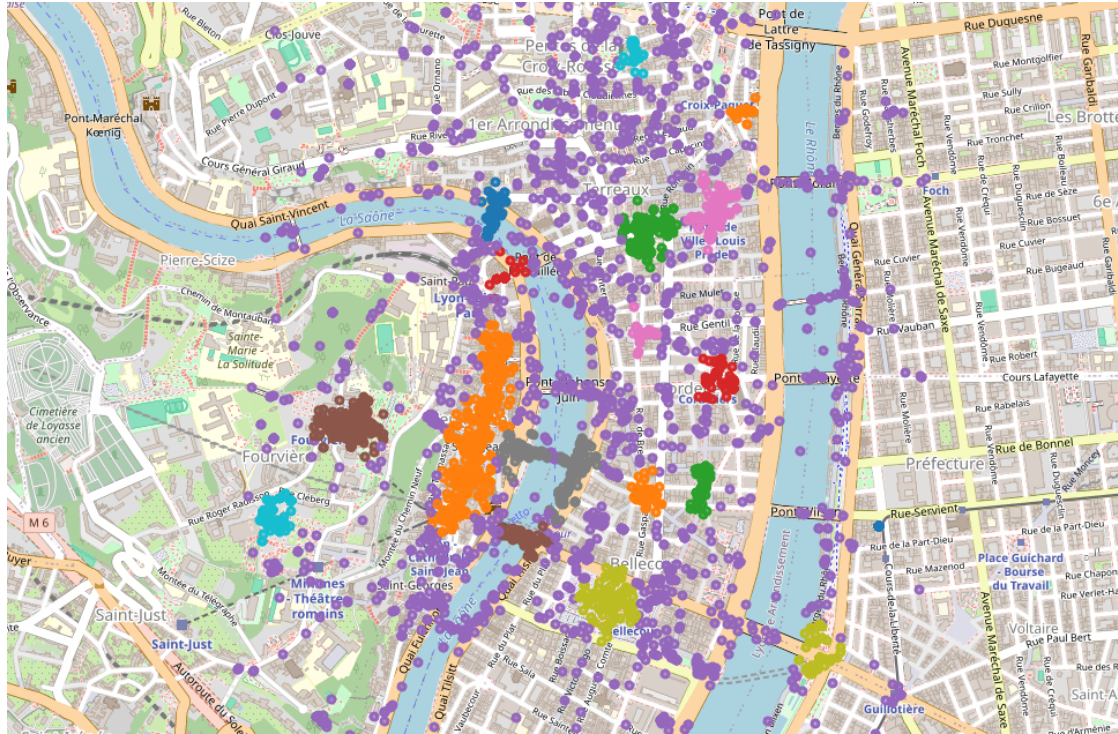


Figure 8: Clustering in the center of Lyon with DBSCAN (Secondary Clusters)

This in-depth clustering allows us to add a few more points of interest:

- The slopes of Croix Rousse - cyan
- Saint-Nizier Church - pink
- Cordelier - red
- Place de la République - green
- Place des Jacobins - orange
- The Saint-Vincent (blue), Pierre Truche (gray), and Bonaparte (brown) footbridges

5.2.3 Conclusion

The algorithm successfully identified geographic points of interest in Lyon and its surroundings at multiple scales. The density-based clustering approach effectively grouped key locations while filtering out noise, providing meaningful insights into urban and suburban structures.

5.3 Agglomerative Clustering

We also explore Agglomerative Clustering, which forms clusters hierarchically. This method was applied both with an automatic number of clusters and a predefined number of 10 clusters.

5.3.1 Hyperparameters

For Agglomerative Clustering, we used the following hyperparameters:

- `n_cluster` default, 10

5.3.2 Results

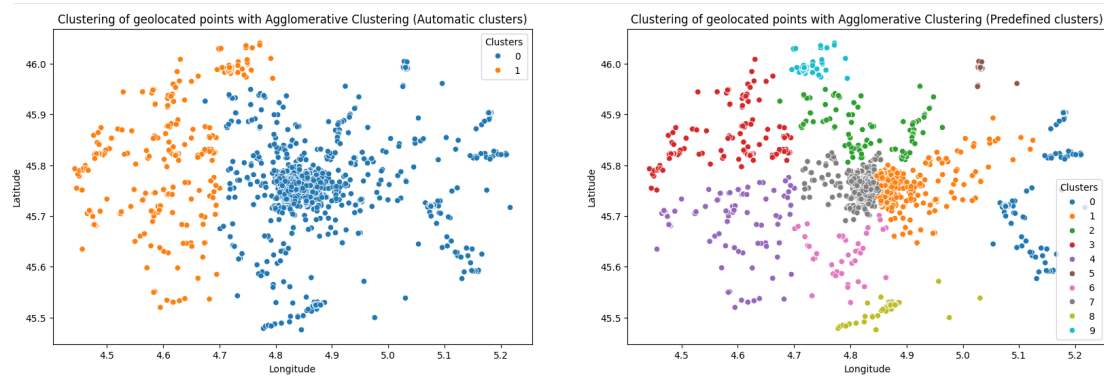


Figure 9: Clustering of geolocated points with Agglomerative Clustering

5.3.3 Conclusion

This method is less effective than DBSCAN for spatial data, as it either oversimplifies the clustering or forces arbitrary divisions.

5.4 Mean-Shift Clustering

Lastly, we evaluate Mean-Shift, which identifies high-density regions without requiring a predefined number of clusters. This method is similar to DBSCAN in its ability to detect clusters based on density, but it operates by iteratively shifting data points toward areas of higher density rather than defining clusters based on a fixed neighborhood radius.

5.4.1 Hyperparameters

For Mean-Shift Clustering, we used the following hyperparameters:

- bandwidth 0.1

5.4.2 Results

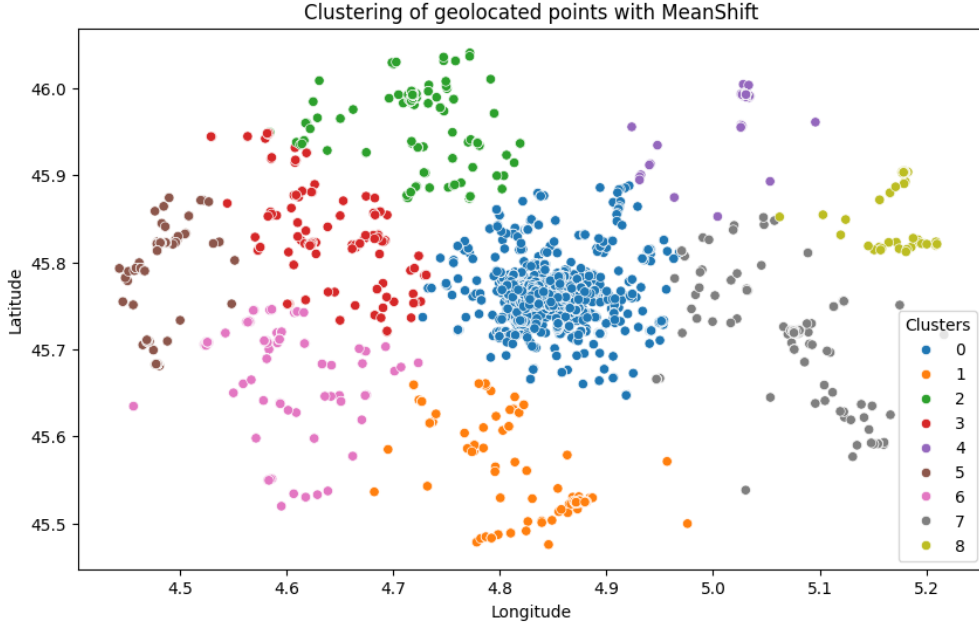


Figure 10: Clusters obtained using Mean-Shift.

5.4.3 Conclusion

Mean-Shift is effective at identifying high-density regions without requiring a predefined number of clusters. However, it tends to misclassify outliers as clusters, which is not ideal for our application, where accurately detecting Points of Interest (PoI) is crucial. Since DBSCAN performs better in this context, we will not explore Mean-Shift further.

6 Conclusion

In this study, we evaluated multiple clustering algorithms to detect Points of Interest (PoI) in Lyon using geospatial data from Flickr. Among the tested methods — **K-Means**, **Mean-Shift**, **Agglomerative Clustering**, and **DBSCAN** — this last one emerged as the most effective.

K-Means struggled due to its assumption of spherical clusters and the need to predefine the number of clusters, which did not align well with the natural distribution of geospatial data. **Mean-Shift**, while useful for detecting high-density regions, tended to misclassify outliers as clusters, limiting its effectiveness for our use case. **Agglomerative Clustering** was also suboptimal, as it either oversimplified the data by forming a single large cluster or imposed arbitrary divisions when the number of clusters was predefined.

In contrast, **DBSCAN** proved to be the best approach for our dataset. Its ability to detect clusters of irregular shapes and varying densities, while effectively isolating outliers, made it particularly well-suited for geographic PoI detection.